

SUNGHO LEE

Automotive Quality Engineer · New Product Launch & Development
Quality (APQP)

Daejeon, Republic of Korea
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Korean (Native) · English (OPIc IM2)

Automotive quality engineer with ~5 years at Tier-1 suppliers (Valeo, Hanon Systems), focused on getting quality right **before SOP**: FMEA-driven launch quality for a new vehicle program, supplier APQP from process validation to line try-out, and data-level root cause analysis of field claims down to IC/EEPROM logic. Now leading supplier development in step with vehicle development gates at Hanon Systems.

CORE STRENGTHS

New Program Launch · APQP Gate-based risk reduction from proto to SOP; FMEA, Control Plan, error-proofing	Customer & Field Quality Led joint investigations with customer and IC maker; KRW 26M claim defended	Automotive Electronics PCBA, Hall IC (TLE4997) EEPROM, FCT/EOLT processes, sensor products
Root Cause Analysis FTA · PDCA · 8D; data-sheet and log-based analysis, reproduction tests	Audit & QMS Led IATF 16949 and Valeo V5000 audits (2023, 2024); supplier process audits	Data & Automation SAP–Apriso inspection interlock; self-built dashboards and tools (VBA, Python, Apps Script)

EXPERIENCE

Hanon Systems 05/2026 – Present
Senior Professional, Parts & Supplier Development (P&SD), Purchasing Division · Daejeon · Full-time

- **Supplier development aligned with vehicle development phases** for heat-exchanger parts (chiller, condenser, radiator): drive quality and delivery readiness at each development gate.
- **JG program chiller development follow-up**: track development progress, check gate-by-gate quality and delivery issues, and assess supplier capacity (CAPA) and processes to confirm SOP readiness.
- **Supplier Communication Seminar (H1 2026)**: planned and hosted for ~200 supplier CEOs, delivering customer (OEM) messages such as Zero Defect and the QCD improvement direction.
- **Supplier quality KPIs**: structured supplier-level quality indicators into a management baseline and shared best-practice improvement cases across suppliers.

Valeo Mobility Korea 01/2025 – 05/2026
Quality Engineer (Site SQA) · Daegu · Full-time · New legal entity and new plant after ADAS spin-off

ROA new program launch — preventive launch quality

- Rated launch risks with FMEA at the design stage: **mixed-part shipment (Middle)** and **connector cosmetic damage (High)**.
- Proposed an **error-proof labeling logic** (label prints only when barcode part number and quantity match); won approval from manufacturing engineering by quantifying the loss (~KRW 5M per mix-up, recurring at least quarterly). Added **connector vision inspection**.
- At proto stage, resolved **bracket noise** with R&D by tightening drawing tolerance (**0.3 → 0.1 mm**) and adding ribs; specified heat-staking and assembly-interference gauges to set clear acceptance criteria.
- Result: FMEA risks lowered to **Low** before SOP; stable early-production assembly and function quality.

New plant QMS & incoming quality

- Built an **SAP–Apriso integrated interlock** that blocks goods receipt until inspection results are entered; re-set key control dimensions, renewed supplier inspection agreements, and digitized ISIR/inspection records to meet Valeo 5 Star requirements.
- Cut in-process scrap rate by **13%** through incoming defect prevention.
- Solved PBT GF30 housing brittleness at an injection-molding supplier (resin residence during machine stops); purging interlock and destructive check lowered FMEA risk from High to Low, rolled out to all molding lines.

Field claim RCA — MDPS steering (sensor product)

- Steering field issue registered as a customer major quality agenda. With no R&D support on site, self-studied the Hall IC data sheet and EEPROM structure, analyzed returned parts (PGSIS-2 tool, oscilloscope reproduction tests).
- Found a Threshold Margin Level written in an unstable band causing parity errors; proved the customer's re-writing voltage drop as root cause in a joint investigation with the customer and IC maker — ~KRW 26M claim cost defended.
- Set up EEPROM data-integrity sampling inspection, limit samples, and inspection standards.

New supplier APQP — SMT/PCBA resourcing (2022.12 – 2023.03)

- Led process audits, PV test selection, FDPR, Cp/Cpk and MSA, and FMEA/Control Plan updates for dual-sourcing a PCBA supplier.
- At line try-out, traced 100% FCT failure to a wrong EEPROM write value (LP filter '100' instead of '011') via data-sheet analysis; added write-then-read 100% verification to the Control Plan — 45% PPM reduction after SOP.

Production & system quality

- Led IATF 16949 and Valeo V5000 (TQ) audits in 2023 and 2024; certification maintained.
- New-mold ISIR approval for 3 injection parts (2024): PFMEA update, capability/MSA review, gauge specs — FDPR process defect rate -10%.
- TAS line top scrap (EOLT hysteresis): identified PCBA flux tilting and housing structure causes — 21% PPM reduction.
- Handled supplier quality issues with global suppliers (Thailand, Slovakia, Germany) in English; recovered quality costs incl. ~KRW 60M and ~KRW 80M claims.

Dokkaebi Co., Ltd.

10/2020 – 03/2021

Product Designer · Gumi · Full-time · Left due to company closure

- Designed an e-kickboard body and safety accessories (wind shield, LED safety system); built prototypes with 3D printing and verified on vehicle.
- Led a national R&D project prototype; coordinated welding/machining vendors for design feasibility and function checks.

DIGEN Co., Ltd.

08/2019 – 12/2019

Process Engineering Intern (Electronics) · Daegu

- Built an Arduino-based USB current-measurement test module to prove the test logic, then led outsourced development of a production 100% automated inspection system.
- Analyzed the Bluetooth communication protocol between sensors to extract valid data and set up real-time monitoring of inspection data.

SKILLS

Quality methods

APQP, PPAP/ISIR, PFMEA, Control Plan, FDPR, SPC (Cp/Cpk), MSA/Gage R&R, FTA, PDCA, 8D, 4M change

Standards & audits

IATF 16949, Valeo V5000 / 5 Star, supplier process audit

Electronics

PCBA/SMT, FCT/EOLT, Hall IC EEPROM analysis (PGSIS-2), oscilloscope

Systems

SAP, Apriso (MES)

Data & automation

Excel VBA, Python (pandas), Google Apps Script, HTML/JS dashboards (Chart.js), Power Automate, Arduino

Working style

Cross-functional with R&D, manufacturing engineering, customers and suppliers; English e-mail and video calls with global suppliers

EDUCATION

Kumoh National Institute of Technology — B.S. Mechanical Engineering (GPA 3.81/4.5)

03/2014 – 02/2020

ADDITIONAL

- Military service: Republic of Korea Navy (2015 – 2016), selected for the Cheonghae Unit anti-piracy escort mission off Somalia.
- OPIc English IM2 (Aug 2025).